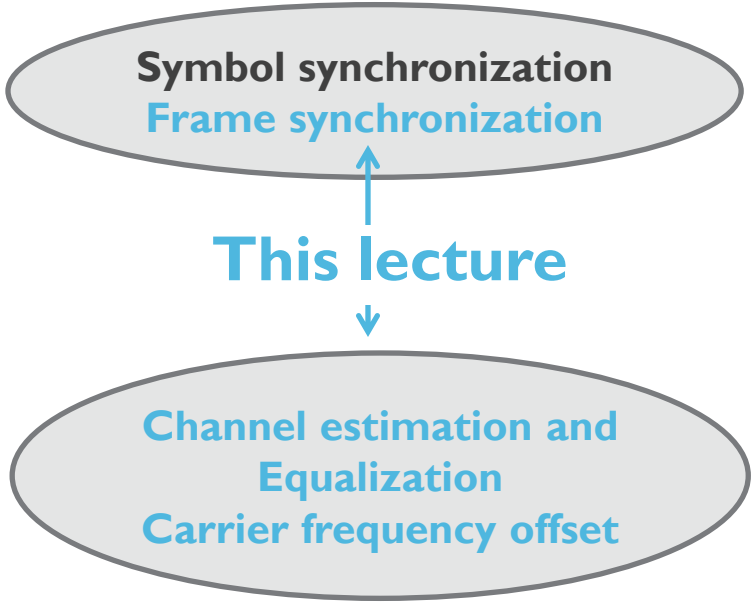
Lecture #11

EE 471C / EE 381K-17 Wireless Communication Lab

From last lecture

- Flat channel model $h_c(t) = \square\delta(t - \square_d)$

  - τ_d → Symbol timing mismatch; Frame timing mismatch
  - α → Amplitude distortion



Preview of Today's Lecture

- Frame synchronization for frequency flat channels
  - Formulate the frame synchronization problem
  - Solve frame synchronization using energy detector or a correlation
- Interlude on least squares
  - Define and compute the least-squares solution for a linear system
- Frequency flat channel estimation
  - Formulate and solve the frequency-flat channel estimation problem
  - Implement detection with channel estimation
- Carrier frequency offset
  - Explain the origin of carrier frequency offset
  - Present a simple carrier frequency offset estimator for flat-fading channels

Frame synchronization

Learning objectives:

- Formulate the frame synchronization problem.
- Solve the frame synchronization problem using a correlator

Frame synchronization system model

- In a frequency-flat channel, after symbol synchronization

$$y[n] = h \sum_{m=-\infty}^{\infty} s[m] g((n-m)T - \tau_{\mathrm{d}}) + v[n]$$

$$\tau_d = dT + \tau_{d,\mathrm{frac}}$$

Symbol synchronization resolves the part of the delay in [0,T), it is not able to resolve delays that are multiples of *T*

- A residual offset of d symbols will remain

$$h = \alpha e^{-\mathrm{j}2\pi f_{\mathrm{c}} \tau_{\mathrm{d}}} \sqrt{E_{\mathrm{x}}}$$

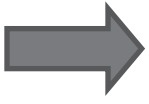
Received signal → $y[n] = hs[n - d] + v[n]$

residual offset

Motivation for frame synchronization

- To reconstruct the transmitted bit sequence it is necessary to know where the symbol stream starts
- Example: consider a 4-PAM system transmiting ASCII symbols

Bits	Symbol
00	-3*a*
01	-*a*
10	+*a*
11	+3*a*



Character	Binary	Symbols
a	01 10 00 01	-*a* +*a* -3*a* -*a*
b	01 10 00 10	-*a* +*a* -3*a* +*a*
c	01 10 00 11	-*a* +*a* -3*a* +3*a*
…	…	…

- Received symbol sequence $\cdots, -a, -a, +a, -3a, -a, \cdots$
- Which character was sent, X or a?

$$\cdots, \underbrace{-a, -a, +a, -3a}_{\text{X}}, -a, \cdots \qquad \cdots, -a, \underbrace{-a, +a, -3a, -a}_{\text{a}}, \cdots$$

Use training data to help with correlation



- Training sequence is known information inserted into the transmission
- Known information is useful for
  - Frame and carrier synchronization …
  - Channel estimation
- Most commercial wireless systems have multiple forms of training
  - Known as training sequences, pilots, reference signals, etc.

Frame synchronization using training sequence

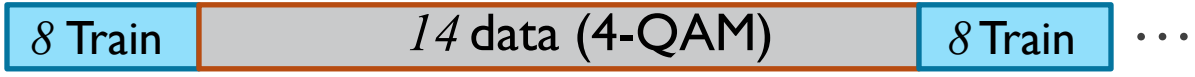
- Suppose $\{t[n]\}_{n=0}^{N_{\text{tr}}-1}$ is the training sequence known at the receiver

- Correlation based detector correlates with the training sequence to compute

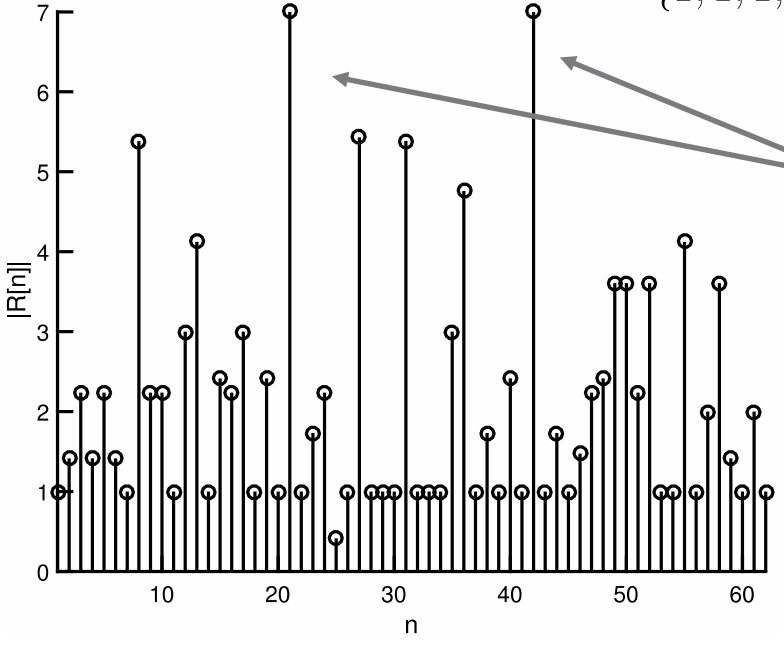
$$R[n] = \sum_{k=0}^{N_{\text{tr}}-1} t^*[k]y[n+k]$$

and solves for $\hat{d} = \arg\max_n |R[n]|$

Example



$\{1, 1, 1, -1, -1, 1, -1\}$ Barker code of length 7



Peaks at 21 and 42

SNR of 5dB

Improving robustness using periodic insertion

- *Periodic insertion* of the training symbols can improve performance and resistance to finding the synchronization sequence in the data

- Correlate over multiple periods

$$\hat{d} = \arg\max_{n} \sum_{p=0}^{P-1} \left| \sum_{k=0}^{N_{\text{tr}}-1} t^*[k] y[n + k + pN_{\text{tot}}] \right|$$

N_{tot} is the length of the frame in symbols, P is the number of periods

Design of training sequences

- Training sequence design depends on the application
- For frame synchronization, sequences should have good auto-correlation properties
  - One example is the Frank sequence

$$t[n] = t[mq+k] = e^{j2\pi rkm/q} \quad 0 \le k, j < q, \ (r,q) = 1$$

where $0 \le n \le q^2 - 1$ and q is any integer

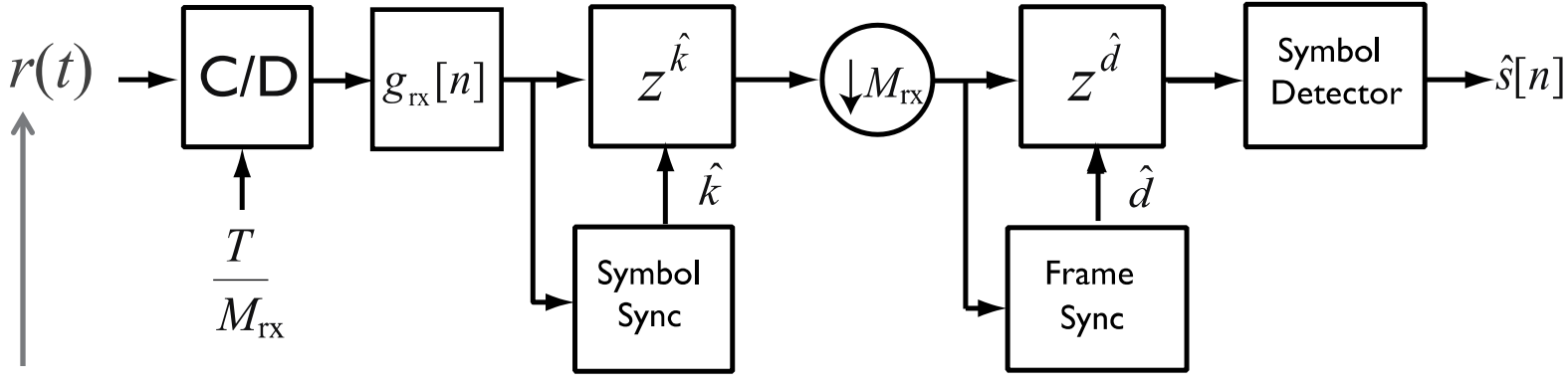
- We can take $q = 4$ and $r = 3$ to obtain a length 16 PSK sequence

  - Another example is the Neuman-Hoffman binary code

$$(-1,-1,-1,-1,-1,-1,\ 1,\ 1,-1,-1,\ 1,-1,\ 1)$$

Receiver block diagram

- Receiver including symbol synchronization and frame synchronization



Complex baseband received signal

Neglecting effects caused by the unknown channel h, to be dealt with next

Frame synchronization

Learning objectives:

- Formulate the frame synchronization problem.
- Solve the frame synchronization problem using a correlator

Interlude on Least Squares and Maximum Likelihood

Learning objectives:

- Define and compute the least-squares solution for a linear system
- Connect least squares and maximum likelihood parameter estimation

System of linear equations

- Consider a system of linear equations written in matrix form

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

  - $\mathbf{A}$: NxM is the known matrix of coefficients called the **data**
  - $\mathbf{x}$: Mx1 is a vector of **unknowns**
  - $\mathbf{b}$: Nx1 is a known vector often called the **observation vector**

Visualization the product

- Expanding

$$\mathbf{Ax} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_M] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_M \end{bmatrix}$$

$$= \sum_{m=1}^{M} \mathbf{a}_m x_m$$

$$= \mathbf{b}$$

- This shows that an exact solution is a linear combination of the columns of $\mathbf{A}$

Minimizing the squared error

- Suppose that $\mathbf{A}$ is square $N = M$ and that $\mathbf{A}$ is invertible (or full rank) then

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

- Suppose that $N > M$ a.k.a. $\mathbf{A}$ is "tall" and $\mathbf{A}$ is full rank (implies that the columns are linearly independent)
  - System is overdetermined
  - Propose to find the solution to minimize the squared error

$$\min \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$$

Finding the minimizer

- Using matrix calculus (differentiating w.r.t. complex conjugate)

$$\begin{aligned}\frac{d}{d\mathbf{x}^c}\|\mathbf{A}\mathbf{x}-\mathbf{b}\|^2 =& \frac{d}{d\mathbf{x}^c}\mathbf{x}^*\mathbf{A}^*\mathbf{A}\mathbf{x} - \frac{d}{d\mathbf{x}^c}\mathbf{x}^*\mathbf{A}^*\mathbf{b} - \frac{d}{d\mathbf{x}^c}\mathbf{b}^*\mathbf{A}\mathbf{x} + \frac{d}{d\mathbf{x}^c}\mathbf{b}^*\mathbf{b} \\ =& \mathbf{A}^*\mathbf{A}\mathbf{x} - \mathbf{A}^*\mathbf{b}.\end{aligned}$$

- Setting equal to zero leads to the **orthogonality condition**

$$\mathbf{A}^*(\mathbf{A}\mathbf{x}-\mathbf{b}) = \mathbf{0}$$

- Simplifying leads to the **normal equations**

$$\mathbf{A}^*\mathbf{A}\mathbf{x} = \mathbf{A}^*\mathbf{b}$$

Least squares solution

- A*A is a square matrix and is invertible because A is full rank, thus

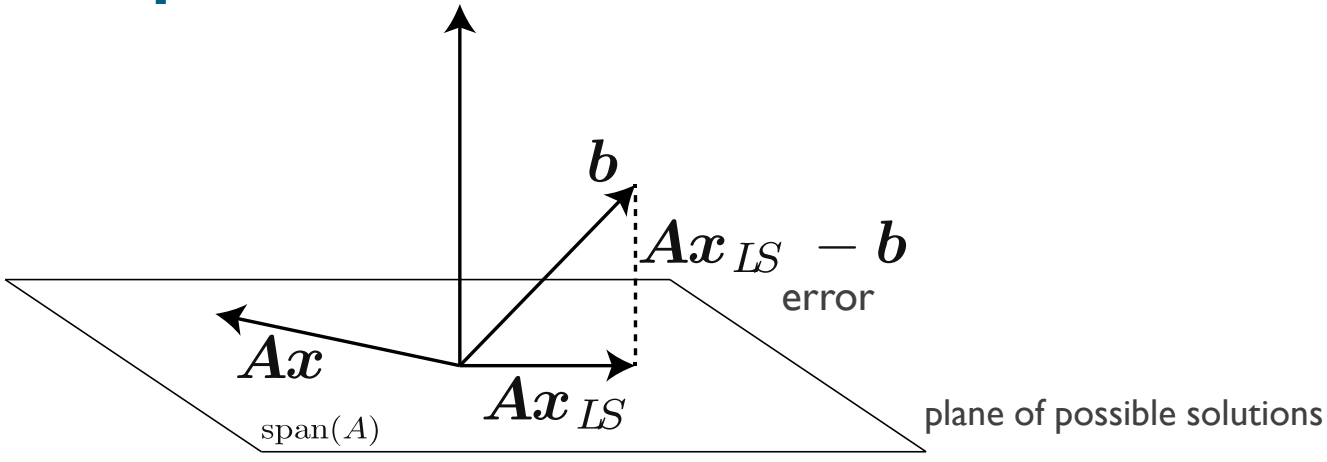
$$\mathbf{x}_{LS} = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*\mathbf{b}$$

- The squared error is

$$\begin{aligned} J(\mathbf{x}_{LS}) &= \|\mathbf{A}\mathbf{x}_{LS} - \mathbf{b}\|^2 \\ &= \mathbf{x}_{LS}^*\mathbf{A}^*(\mathbf{A}\mathbf{x}_{LS} - \mathbf{b}) - \mathbf{b}^*(\mathbf{A}\mathbf{x}_{LS} - \mathbf{b}) \\ &= \mathbf{b}^*\mathbf{b} - \mathbf{b}^*\mathbf{A}\mathbf{x} \end{aligned}$$

Measures the performance of the estimator

Geometric interpretation



- $\mathbf{A(A^*A)^{-1}A^*b}$ is the projection of b onto the space spanned by the columns of A
- The error is $\mathbf{(A(A^*A)^{-1}A^* - I)b}$ is the projection of $\mathbf{b}$ onto the orthogonal complement of the rows of $\mathbf{A}$

Maximum likelihood estimation

- Suppose there is measurement noise, modeled as N(0,1)

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{v}$$

- Use maximum likelihood to estimate x
- Give a value of x called $\bar{\mathbf{x}}$ the likelihood function is (A known also)

This result is only true for IID Gaussian noise

$$f_{\mathbf{y}|\mathbf{A},\mathbf{x}}(\mathbf{y}|\mathbf{A},\bar{\mathbf{x}}) = \frac{1}{\pi^N} e^{-(\mathbf{y}-\mathbf{A}\bar{\mathbf{x}})^*(\mathbf{y}-\mathbf{A}\bar{\mathbf{x}})}$$

- Differentiating and setting the result equal to zero gives

$$\mathbf{A}^*(\mathbf{y} - \mathbf{A}\bar{\mathbf{x}}) = \mathbf{0} \quad \Rightarrow \quad \mathbf{x}_{\mathrm{ML}} = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*\mathbf{y}$$

Maximum likelihood estimate corresponds to the least squares solution neglecting noise

Interlude on Least Squares and Maximum Likelihood

Learning objectives:

- Define and compute the least-squares solution for a linear system
- Connect least squares and maximum likelihood parameter estimation

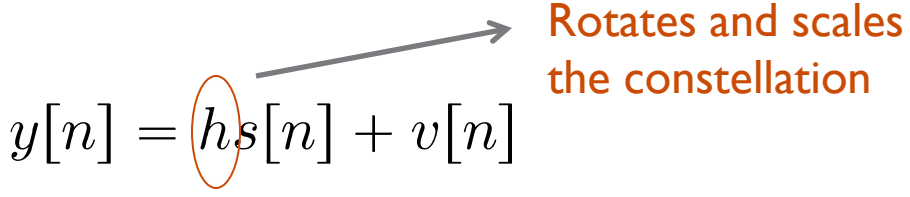
Frequency flat channel estimation

Learning objectives:

- Formulate and solve the frequency-flat channel estimation problem
- Implement detection with channel estimation

Frequency flat channel estimation

- Suppose that symbol and frame synchronization are solved then



- Sampling and removing delay leaves distortion due to h
- Some modulation schemes are robust to such distortion (e.g., differential QPSK)
- For general linear modulation, need channel estimation and equalization

Problem formulation

- Let complex scalar h be the unknown channel

$$y[n] \quad = \quad hs[n] \; + \; v[n]$$

- Channel estimation
  - Exploit known training $\{t[n]\}_{n=0}^{N_{\rm tr}-1}$ to estimate the channel

- Write received signal in vector form using known training
  - Noise is IID therefore the least squares estimator is the maximum likelihood estimator

Channel estimation

- Signal model

$$\underbrace{\begin{bmatrix} y[0] \\ y[1] \\ \vdots \\ y[N_{\text{tr}}-1] \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} t[0] \\ t[1] \\ \vdots \\ t[N_{\text{tr}}-1] \end{bmatrix}}_{\mathbf{t}} h + \underbrace{\begin{bmatrix} v[0] \\ v[1] \\ \vdots \\ v[N_{\text{tr}}-1] \end{bmatrix}}_{\mathbf{v}} \quad \Rightarrow \quad \boxed{\mathbf{y} = \mathbf{t}h + \mathbf{v}}$$

- LS solution is

$$\hat{h} = (\mathbf{t}^*\mathbf{t})^{-1}\mathbf{t}^*\mathbf{y} = \frac{\sum_{n=0}^{N_{\text{tr}}-1} t^*[n]y[n]}{\sum_{n=0}^{N_{\text{tr}}-1} t^*[n]t[n]}$$

Correlator – can jointly perform frame synchronization and channel estimation

Corrects for scaling

ML detector

- ML detector with channel estimation
  - To correct for h simply include estimated channel in ML detector

$$\hat{s}[n] = \arg\min_{s \in \mathcal{C}} |y[n] - \hat{h}s|^2$$

  - Channel estimate becomes an input into the detector

- Using estimated $\hat{h}$ introduces an additional source of errors

Equalization

- Remove the effect of the channel prior to detection

$$\arg\min_{s\in\mathcal{C}} |y[n]-\hat{h}s|^2 = \arg\min_{s\in\mathcal{C}} |\hat{h}|^{-1}\left|\frac{y[n]}{\hat{h}}-s\right|^2 = \arg\min_{s\in\mathcal{C}}\left|\frac{y[n]}{\hat{h}}-s\right|^2$$

Equalization

- A standard detector can be applied to the equalized signal

Equalization

- The probability of symbol error can be computed treating channel estimation error as noise
- Equalized received signal

$$\hat{s}[n] = \frac{1}{\hat{h}}(\hat{h} + \hat{h}_{\mathrm{e}})s[n] + \frac{1}{\hat{h}}v[n]$$
$$= s[n] + \frac{\hat{h}_{\mathrm{e}}}{\hat{h}}s[n] + \frac{1}{\hat{h}}v[n].$$

Creates interference, treat as noise

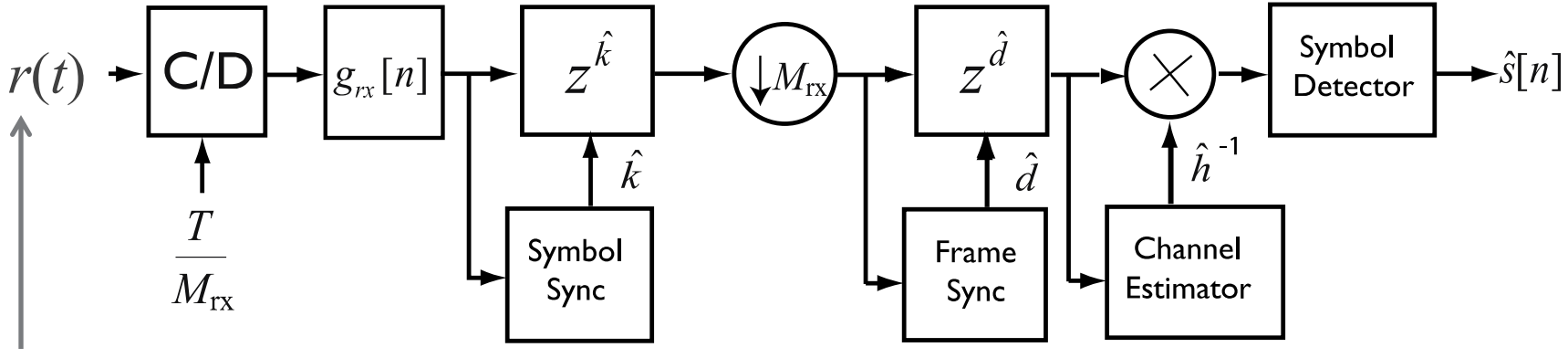
- Impact of channel estimation in SINR

Replace with error variance

$$\mathrm{SINR}_{\hat{h}_{\mathrm{e}}} = \frac{|\hat{h}|^2}{|\hat{h}_{\mathrm{e}}|^2 + \sigma_v^2} \quad \text{or} \quad \mathrm{SINR} = \frac{|\hat{h}|^2}{\mathbb{E}\left[|\hat{h}_{\mathrm{e}}|^2\right] + \sigma_v^2}$$

Receiver block diagram

- Receiver with symbol synchronization, frame synchronization, and channel estimation



Frequency flat channel estimation

Learning objectives:

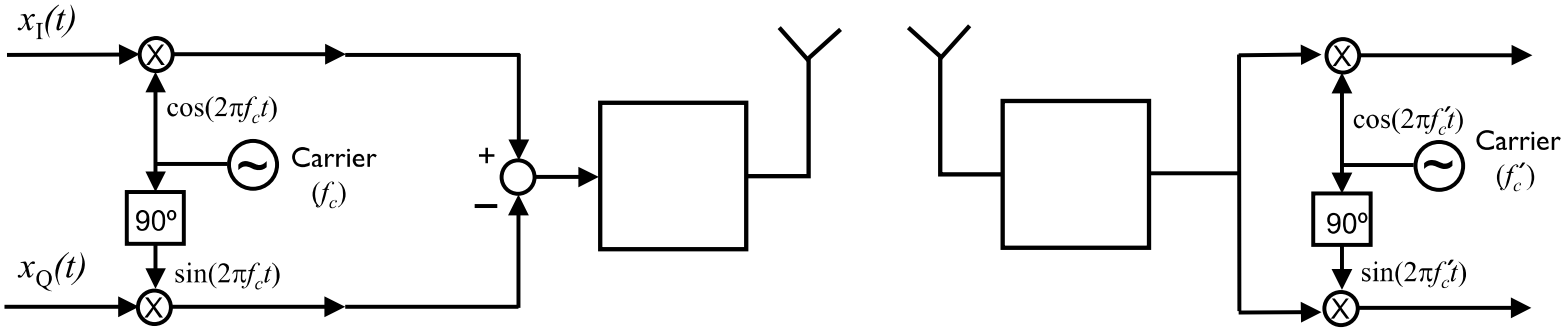
- Formulate and solve the frequency-flat channel estimation problem
- Implement detection with channel estimation

Frequency Offset

Learning objectives:

- Explain origin of frequency offset and discuss nature of the impairment
- Analyze a simple offset estimation algorithm based on sending a sinusoid

Downconversion



- Consider the received signal after downconversion

- What if only f_c' not f_c is known at the receiver?
  - The result is carrier frequency offset (CFO)

Downconversion

- Let $f_e = f'_c - f_c$

$$
\begin{aligned}
r_{\mathrm{p}}(t)\cos(2\pi f'_{\mathrm{c}}t) &= \frac{1}{2}r_{\mathrm{i}}(t)\cos(2\pi f_{\mathrm{e}}t) \\
&\quad - \frac{1}{2}r_{\mathrm{i}}(t)\cos(2\pi(f_{\mathrm{c}}+f'_{\mathrm{c}})t) - \frac{1}{2}r_{\mathrm{q}}(t)\sin(2\pi(f_{\mathrm{c}}+f'_{\mathrm{c}})t) \\
-r_{\mathrm{p}}(t)\sin(2\pi f'_{\mathrm{c}}t) &= \frac{1}{2}r_{\mathrm{q}}(t)\sin(2\pi f_{\mathrm{e}}t) \\
&\quad + \frac{1}{2}r_{\mathrm{q}}(t)\cos(2\pi(f_{\mathrm{c}}+f'_{\mathrm{c}})t) + \frac{1}{2}r_{\mathrm{i}}(t)\sin(2\pi(f_{\mathrm{c}}+f'_{\mathrm{c}})t)
\end{aligned}
$$

- Low pass filtering removes high frequency components

$$
\begin{aligned}
r'(t) &= e^{\mathrm{j}2\pi f_{\mathrm{e}}t}\left(r_{\mathrm{i}}(t) + \mathrm{j}r_{\mathrm{q}}(t)\right) \\
&= e^{\mathrm{j}2\pi f_{\mathrm{e}}t}r(t).
\end{aligned}
$$

derotate to remove

- Correction of f_{e} is carrier frequency synchronization

Mathematical relationship

- Assuming the offset is small, the front-end bandwidth is sufficiently wide, and the pulse shape is concentrated over a few symbols then

$$y(t) = e^{\mathrm{j}2\pi f_{\mathrm{e}} t} \int r(t-\tau) g_{\mathrm{rx}}(\tau) \mathrm{d}\tau.$$

after the convolution

- In discrete time, including noise and with $\epsilon = f_{\mathrm{e}} T$

$$y[n] = e^{\mathrm{j}2\pi\epsilon n} h \sum_{m=-\infty}^{\infty} s[m] g((n-m)T - \tau_{\mathrm{d}}) + v[n]$$

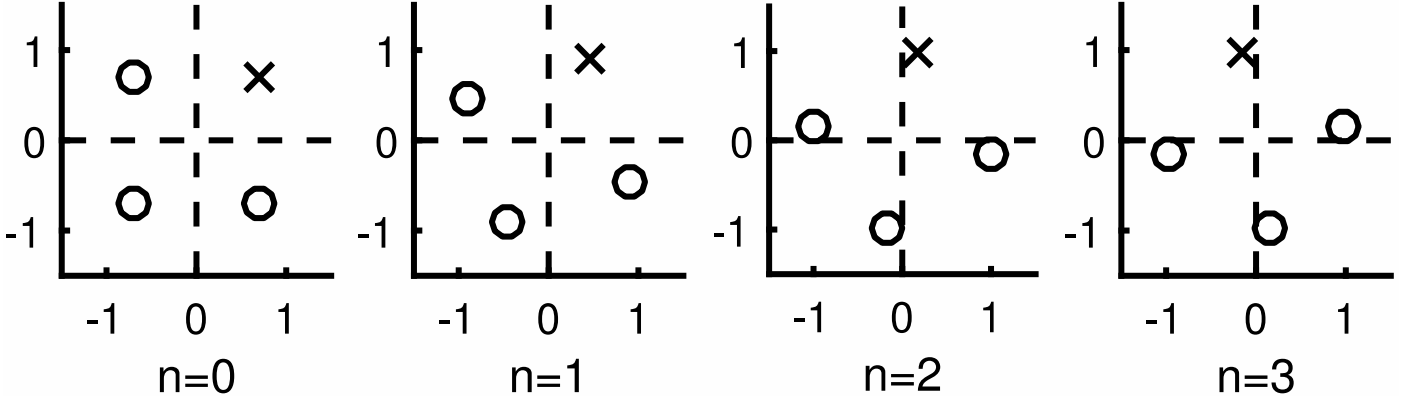
  - Rotation occurs after the convolution
  - Impacts channel estimation and thus equalization

Visualizing the frequency offset effect

- Special case of flat fading channel

$$y[n] = e^{j2\pi\epsilon n} hs[n] + v[n]$$

  - ϵ is generally small but unknown
  - Rotates constellation by $e^{j2\pi\epsilon n}$



Frequency offset synchronization

- Frequency offset is a severe impairment
  - Even a small amount of offset leads to significant degradation
- Frequency offset synchronization is challenging
  - Offset occurs after the convolution with the unknown channel
  - Impacts channel estimation and frame synchronization
- Methods for offset correction
  - Exploit structure in the received signal
  - Create and exploit structure using a known training signal

Frequency offset estimation using a sinusoid

- Suppose that frame synchronization has been solved
- Choose as a training sequence a complex sinusoid

$$t[n] = \exp(\mathrm{j}\omega n) \; for \; n = 0, 1, \ldots, N_{\mathrm{tr}} - 1$$

- The received signal during training is

$$y[n] = e^{\mathrm{j}2\pi\epsilon n} e^{\mathrm{j}\omega n} h + v[n]$$

- Removing the known offset

$$e^{-\mathrm{j}\omega n} y[n] = e^{\mathrm{j}2\pi\epsilon n} h + e^{-\mathrm{j}\omega n} v[n]$$

This becomes a classic problem of estimating a complex sinusoid with unknown amplitude, frequency and phase (amplitude and phase from the channel)

Approximate least squares solution

- Approximate additive noise as phase noise

$$e^{-\mathrm{j}\omega n} y[n] \approx |h| e^{\mathrm{j}2\pi\epsilon n + \theta + \nu[n]}$$

Phase of h — Gaussian noise in phase term

- Looking at the phase difference between two adjacent samples

$$\mathrm{phase}(e^{\mathrm{j}\omega n} y^*[n] e^{-\mathrm{j}\omega(n+1)} y[n+1]) = 2\pi\epsilon + \nu[n+1] - \nu[n]$$

- Aggregating observations from n=1,...,N_{tot}-1

$$\mathbf{p} = 2\pi\epsilon \mathbf{1} + \boldsymbol{\nu}$$

- **LS solution**

$$\hat{\epsilon} = \frac{1}{2\pi}(\mathbf{p}^*\mathbf{p})^{-1}\mathbf{1}^*\mathbf{p}$$

Just the average phase difference

$$= \frac{1}{2\pi(N_{\mathrm{tr}}-1)} \sum_{n=1}^{N_{\mathrm{tr}}-1} \mathrm{phase}(e^{\mathrm{j}\omega} y^*[n] y[n+1])$$

Frequency Offset

Learning objectives:

- Explain origin of frequency offset and discuss nature of the impairment
- Analyze a simple offset estimation algorithm based on sending a sinusoid